

Advanced (Habanero)

Q1. Factor $x^2 - 4$

- (A) $x^2 - 2^2$
- (B) $(x + 2)(x - 2)$
- (C) $(x - 2)^2$
- (D) $x^2 + 4$
- (E) $(x + 1)(x - 1)$

Q2. What is the factored form of $25a^2 - 9b^2$?

- (A) $5a - 3b$
- (B) $(5a + 3b)(5a - 3b)$
- (C) $(5a - 3b)^2$
- (D) $(25a^2 + 9b^2)$
- (E) $5a + 3b$

Q3. Factor $16x^2 - y^2$

- (A) $(4x - y)(4x + y)$
- (B) $16x^2 + y^2$
- (C) $(4x)^2 - (y)^2$
- (D) $4x - y$
- (E) $x^2 - y^2$

Q4. What is the factored form of $x^2 - 16$?

- (A) $x^2 + 16$
- (B) $(x + 4)(x - 4)$
- (C) $x^2 - 4^2$
- (D) $(x - 4)^2$
- (E) $x^2 + 4^2$

Q5. Factor $49m^2 - 4n^2$

- (A) $(7m - 2n)(7m + 2n)$
- (B) $49m^2 + 4n^2$
- (C) $(7m)^2 - (2n)^2$
- (D) $7m - 2n$
- (E) $7m + 2n$

Q6. What is the factored form of $36a^2 - 64b^2$?

- (A) $(6a + 8b)(6a - 8b)$
- (B) $36a^2 + 64b^2$
- (C) $(6a)^2 - (8b)^2$
- (D) $(36a^2 - 8b^2)$
- (E) $6a - 8b$

Q7. Factor $9x^2 - 9y^2$

- (A) $9(x + y)(x - y)$
- (B) $3x - 3y$
- (C) $9x^2 + 9y^2$
- (D) $(3x)^2 - (3y)^2$
- (E) $3x + 3y$

Q8. What is the factored form of $4x^2 - 9y^2$?

- (A) $(2x + 3y)(2x - 3y)$
- (B) $2x + 3y$
- (C) $4x^2 + 9y^2$
- (D) $(2x)^2 - (3y)^2$
- (E) $2x - 3y$

Open Ended Questions (FRQ)

Q9. Factor the expression $x^2 - 20x + 100 - y^2$ and simplify.

Q10. Factor the expression $16x^2 - 9$ and solve for x when the expression equals zero.

Chef's Tips

- Remind students of the difference of squares formula $a^2 - b^2 = (a + b)(a - b)$.
- Emphasize that for factoring expressions like $a^2 - b^2$, it's crucial to identify a and b correctly.
- Encourage students to double-check their factored forms by expanding them.